

Smart Forgotten Appliance System Using ESP32 with PIR Motion Sensor, ACS712 Current Sensor and WebSocket-Based Real-Time Web Dashboard

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Abstract

Energy wastage caused by appliances such as lights, ceiling fans, heaters, and air conditioners that are unintentionally left switched on in unoccupied rooms contributes significantly to rising household and industrial electricity consumption. This paper proposes a low-cost, intelligent embedded solution called the Smart Forgotten Appliance System, designed using the ESP32 microcontroller integrated with a Passive Infrared (PIR) motion sensor for human occupancy detection, an ACS712 Hall-effect current sensor for real-time load current and power measurement, a 5V relay for automated appliance switching, an active buzzer for multi-stage audible warning, and a 0.96-inch SH1106 OLED display for local status indication. A WebSocket-based real-time web dashboard hosted on the ESP32 enables bi-directional, full-duplex communication with sub-100 millisecond latency, allowing the user to monitor instantaneous current, power consumption, motion status, and appliance state remotely. The proposed system implements a three-stage intelligent decision logic: a 10-second warning phase that activates the buzzer when no motion is detected while the appliance is still drawing current, a 20-second auto-off mechanism that switches off the appliance if motion remains absent, and a 2-minute safety cutoff that turns off the load during prolonged continuous operation to prevent overheating and wastage. Experimental validation was performed using a 9W LED bulb as the test load, and the ACS712 module achieved an accuracy of ± 5 mA in current measurement, while the relay-based switching demonstrated reliable response. The integrated dashboard provided seamless real-time visualization. The proposed framework offers a scalable, energy-efficient, and affordable foundation for next-generation smart home automation and Internet of Things (IoT) based energy conservation applications.

Keywords: *Smart Forgotten Appliance System, IoT Energy Conservation, ESP32, PIR Motion Sensor, ACS712 Current Sensor, WebSocket Communication, OLED Display, Embedded Systems, Home Automation, Energy Monitoring.*

1. Introduction

The exponential growth of electrical and electronic appliances in residential, commercial, and industrial environments has resulted in a sharp increase in global electricity consumption. According to recent international energy reports, a substantial portion of domestic electricity is wasted due to user negligence, such as leaving lights, fans, heaters, or air-conditioning units running in rooms that are no longer occupied. In developing economies such as India, where per-capita energy demand is rising rapidly, even small efficiency improvements at the household level can produce significant cumulative savings. Traditional manual switching and timer-based automation methods are insufficient because they do not respond to dynamic human presence and lack remote monitoring capabilities. With the advent of low-cost Internet of Things (IoT) microcontrollers, wireless communication protocols, and energy-monitoring sensors, it has now become feasible to design intelligent appliance-control systems that combine occupancy detection, real-time power measurement, and remote dashboard supervision.

This paper presents a Smart Forgotten Appliance System built around the ESP32 microcontroller, which offers integrated dual-core processing, Wi-Fi, and Bluetooth at very low cost. The system utilizes a PIR motion sensor to detect occupancy, an ACS712 Hall-effect current sensor to continuously measure load current and compute instantaneous power, a 5V relay module for automated load switching, a buzzer for multi-stage audible alerts, and a 0.96" SH1106 OLED for local status display. A WebSocket-based web dashboard hosted directly on the ESP32 ensures sub-100 millisecond, bi-directional communication. The main contributions of this work are: (i) a three-stage decision-making algorithm (10-second warning, 20-second auto-off, 2-minute safety cutoff), (ii) integration of current sensing with motion detection to avoid false triggers, (iii) a low-latency real-time dashboard, and (iv) a low-cost, scalable platform validated experimentally using a 9W LED bulb load.

2. Literature Review

Numerous research works have explored IoT-based smart home automation, occupancy detection, and energy monitoring. The following section reviews five representative studies that motivate the design of the proposed Smart Forgotten Appliance System.

Kim et al. (2020) proposed an IoT-based smart home automation framework using cloud connectivity and centralized control of multiple appliances. The system primarily relied on user-initiated commands and a cloud server for decision making, which introduced latency and a strong dependency on internet availability. Although effective for general automation, the framework did not address autonomous detection of forgotten appliances or local current sensing.

Sharma and Kumar (2021) developed an ESP32-based home automation system using Wi-Fi and a smartphone application to control appliances such as fans and lights. Their system showed the cost-effectiveness and versatility of the ESP32 platform but lacked occupancy sensing and energy consumption monitoring, leaving the energy-wastage problem unresolved.

Singh (2019) studied PIR-based occupancy detection systems for indoor lighting automation. The work demonstrated that PIR sensors are inexpensive and reliable for human motion detection within

short ranges but emphasized the issue of false negatives when occupants remain stationary, suggesting the need for combining PIR data with other indicators such as current draw.

Park and Lee (2020) implemented an HTTP-based motion-triggered appliance control system using a microcontroller, PIR sensor, and relay. While the HTTP polling mechanism allowed remote control, it suffered from high communication overhead and noticeable latency, making real-time monitoring inefficient compared to WebSocket-based architectures.

Patel (2022) proposed an ACS712-based current monitoring framework for analyzing household energy consumption. The work validated the accuracy of the ACS712 Hall-effect sensor for low-current AC loads but did not integrate occupancy sensing or automated switching, limiting its applicability for autonomous energy-conservation systems.

Reference	Methodology	Advantages	Limitations
Kim et al. (2020)	IoT cloud-based home automation	Centralized control, multi-device	Cloud dependency, high latency
Sharma & Kumar (2021)	ESP32 Wi-Fi home automation	Low-cost, smartphone app	No occupancy or energy sensing
Singh (2019)	PIR-based occupancy detection	Simple, low-cost	False negatives, no current data
Park & Lee (2020)	HTTP motion-triggered control	Remote relay control	High latency, communication overhead
Patel (2022)	ACS712 current monitoring	Accurate energy measurement	No automation, no motion sensing

Table 1: Summary of Existing Models

3. Problem Statement

Traditional appliance control mechanisms in residential, commercial, and institutional environments rely on manual switching, where the user is solely responsible for turning appliances ON and OFF. Users frequently forget to switch OFF appliances such as bulbs, fans, and chargers when leaving a room, leading to unnecessary energy consumption. These systems suffer from several critical limitations:

- Users have no automated way to identify whether an appliance has been left ON after they leave the room, leading to unnoticed and prolonged energy wastage.
- Manual switching mechanisms rely entirely on user discipline, which is often inadequate and inconsistent across different users and environments.
- There is no mechanism for warning the user before automatically switching OFF the appliance, resulting in either abrupt shutdowns or persistent wastage when no automation is present.
- Homeowners and administrators lack a centralized, real-time view of appliance status, current consumption, and energy usage to make informed energy management decisions.
- Conventional ON/OFF switching systems cannot generate statistical insights on appliance usage patterns, power consumption, or cumulative energy data.

- Existing low-cost smart home solutions often lack real-time synchronization between the physical appliance state, local status displays, and remote monitoring interfaces due to high-latency HTTP polling.
- Physical override buttons are ineffective in this context, as the appliance is only forgotten when the user is absent, making a remote override mechanism essential.
- Existing systems do not provide a fail-safe safety timeout that ensures the appliance is turned OFF after a predefined maximum duration, even when override mode is active.

4. Objectives

The primary objectives of the Smart Forgotten Appliance System are as follows:

1. To design and implement a low-cost, IoT-based intelligent appliance control system using the ESP32 microcontroller for automatic energy conservation.
2. To detect human presence in real-time using a PIR (Passive Infrared) motion sensor and use it as the primary trigger for appliance ON/OFF decisions.
3. To measure real-time current, power, and cumulative energy consumption of the connected appliance using the ACS712 Hall-effect current sensor.
4. To control the AC bulb load through a relay module based on presence detection and configurable timing parameters.
5. To implement a multi-stage warning mechanism using an active buzzer that alerts the user before automatic shutdown, providing a window for response.
6. To provide real-time local status visibility through a 128×64 OLED display showing bulb state, motion status, current, power, energy, and ON-time.
7. To develop a WebSocket-based responsive web dashboard for remote monitoring and override control, accessible from any device connected to the same network.
8. To achieve WebSocket communication latency below 100 milliseconds for instant real-time synchronization between the ESP32 and the web dashboard.
9. To implement a three-stage decision logic with a 10-second warning trigger, 20-second automatic shutdown, and a 2-minute safety cutoff that operates regardless of override mode.
10. To design a scalable system architecture that supports future enhancements including cloud integration (Firebase/Blynk), mobile application development, multi-appliance control, voice assistant integration, and machine learning-based adaptive behavior.

5. Proposed System

The proposed Smart Forgotten Appliance System integrates occupancy detection, current sensing, automated switching, audible alerting, local display, and remote real-time monitoring into a single ESP32-based embedded platform. The architecture is organized into four functional layers: (i) the sensing layer comprising the PIR motion sensor and the ACS712 current sensor; (ii) the processing layer formed by the ESP32 microcontroller, which performs sensor data acquisition, threshold evaluation, and execution of the three-stage decision algorithm; (iii) the actuation layer composed of the 5V

relay module for load switching and the active buzzer for warning generation; and (iv) the communication and visualization layer consisting of the SH1106 OLED display and the WebSocket-based web dashboard.

The PIR sensor continuously monitors human presence in the controlled area, while the ACS712 sensor samples the load current at a rate of 1 kHz to compute the RMS current and instantaneous power. The ESP32 combines these inputs and applies the three-stage decision logic to determine the appliance state. The real-time dashboard, hosted directly on the ESP32 web server, uses WebSockets to push the latest values to connected clients with sub-100 ms latency, providing a smooth user experience without periodic polling.

This modular system architecture enables future scalability to multiple appliances, integration with cloud platforms, and addition of machine-learning based prediction modules without requiring fundamental redesign.

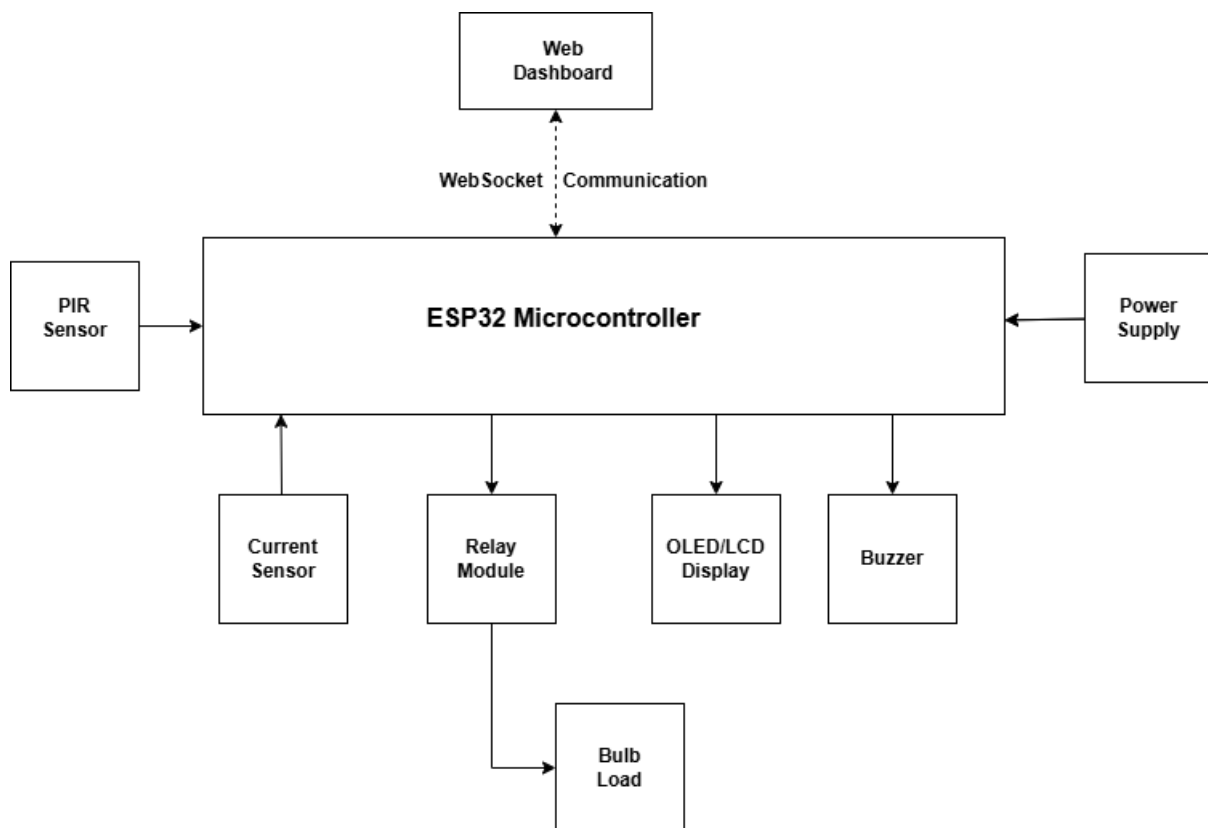


Figure 1: Proposed System Block Diagram

6. Hardware Architecture and Implementation

The hardware implementation is based on widely available and inexpensive components, ensuring that the system can be replicated for real-world deployment. The ESP32 DevKit V1 serves as the central processing and communication unit. The PIR sensor and ACS712 module provide the sensing data,

while the relay module controls the connected appliance load. The OLED display and buzzer provide local feedback, and the Wi-Fi radio of the ESP32 enables WebSocket-based remote dashboard access.

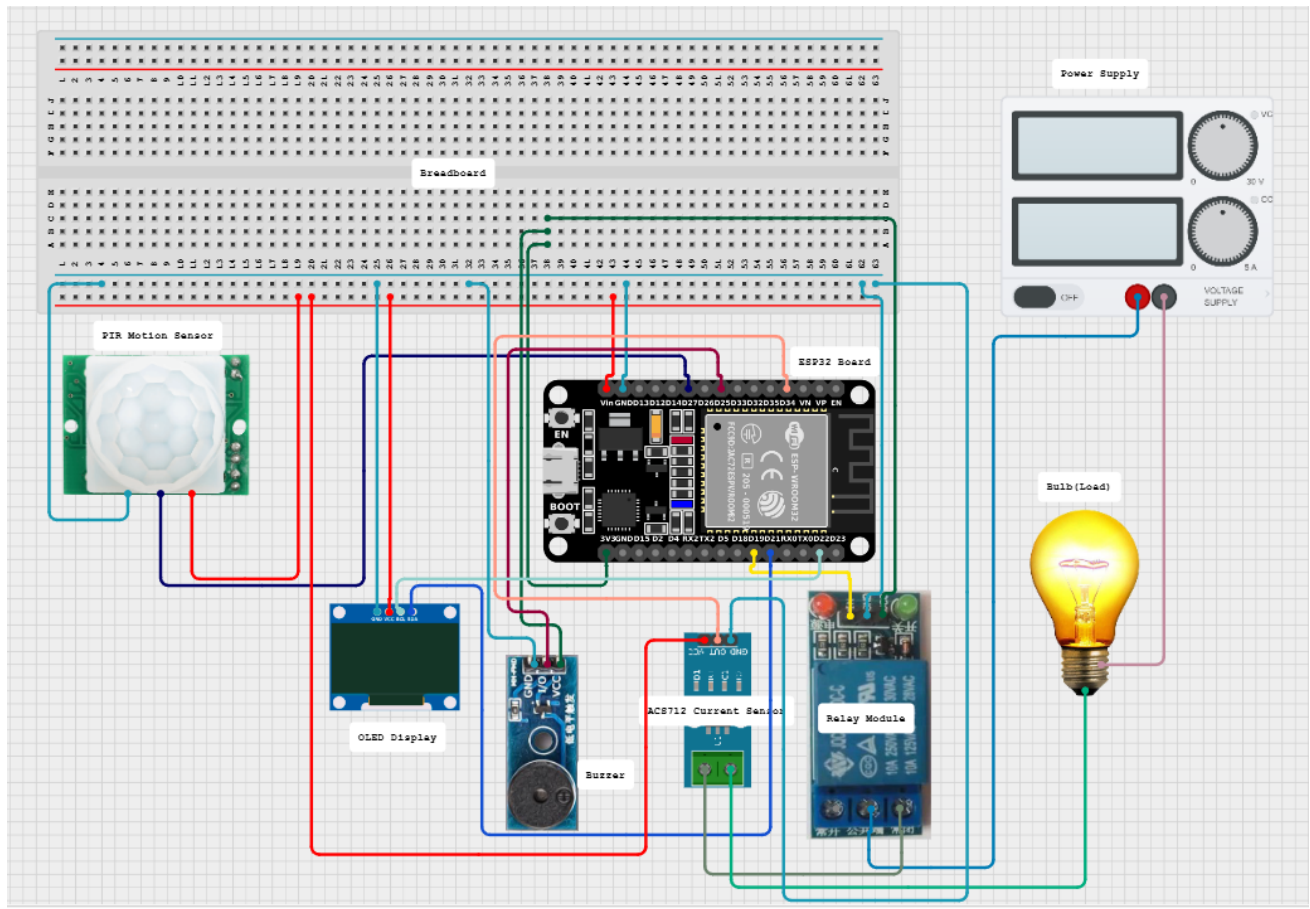


Figure 2: Circuit Diagram of Proposed System

Component	ESP32 Pin
PIR Sensor OUT	GPIO27
ACS712 OUT (Analog)	GPIO34 (ADC1_CH6)
Relay IN	GPIO19
Buzzer	GPIO25
OLED SDA	GPIO21
OLED SCL	GPIO22

Table 2: Pin Connections

6.1. Hardware Components

Table 3 presents the hardware components used in the Smart Forgotten Appliance System along with their specifications and functions.

Component	Specification	Function
ESP32 Microcontroller	Dual-core 240 MHz, 4 MB Flash, 520 KB SRAM,	Central processing unit; handles sensor reading, decision logic, relay

	802.11 b/g/n Wi-Fi, Bluetooth 4.2, 12-bit ADC	control, energy calculation, WebSocket server, and web dashboard hosting
PIR Motion Sensor (HC-SR501)	5V DC operation, digital output (HIGH/LOW), 3–7 meter detection range, 110° detection angle, adjustable sensitivity and time delay	Detects human presence through passive infrared radiation; provides the primary input for inactivity-based decision logic
ACS712 Current Sensor	Hall-effect based, 5A range, 185 mV/A sensitivity, 2.5V offset voltage, 2.1 kV galvanic isolation, analog output	Measures real-time AC current consumption of the appliance load; enables power and energy calculation
Relay Module (5V)	5V DC trigger, 1-channel, active LOW logic, rated 10A at 250V AC, optocoupler isolated	Acts as electrical switch controlled by ESP32; safely switches the AC bulb load ON and OFF based on system decisions
Active Buzzer	3.3V/5V DC operation, active type with internal oscillator, audible output ~85 dB, GPIO-controlled	Produces audible warning alert during the 10–13 second window before automatic shutdown to give the user a chance to respond
OLED Display (SH1106)	128×64 pixels, monochrome, I2C interface, 3.3V/5V compatible, 0.96-inch screen	Displays real-time system status including bulb state, motion status, current, power, energy consumption, ON-time, and override mode locally
AC Bulb (Load)	9 W LED bulb, 230 V AC, E27 base	Demonstration appliance load representing forgotten devices in residential and commercial environments
Wi-Fi Network	IEEE 802.11 b/g/n, 2.4 GHz band, WPA2-PSK security	Connects ESP32 to the local network for real-time WebSocket-based dashboard communication
Power Supply	5V USB, 1A minimum rating	Provides stable power to ESP32 and all connected components (PIR, ACS712, OLED, buzzer, relay)
Breadboard & Jumper Wires	Standard 830-tie breadboard, male-to-male and male-to-female jumper wires	Provides physical interconnection between ESP32 and all peripheral components for prototyping

Table 3. Hardware Specifications

7. Methodology

7.1 Sensing and Initialization Stage

The sensing stage is triggered when the system is powered ON. The ESP32 initializes all peripherals including the PIR motion sensor, ACS712 current sensor, relay module, buzzer, OLED display, and the WiFi connection. The bulb is set to the default ON state by activating the relay (active LOW). The HTTP server is started on port 80 to serve the dashboard HTML page, and the WebSocket server is started on port 81 for real-time bidirectional communication. The PIR sensor continuously monitors human presence by detecting changes in infrared radiation, while the ACS712 sensor measures the real-time current flowing through the connected appliance load. The OLED display and web dashboard are immediately updated to reflect the initial system state.

7.2 Decision Logic Stage

The ESP32 executes a three-stage decision logic that continuously evaluates the inactivity duration and the bulb operating time. Based on the PIR signal and timer values, the system classifies the state into one of three modes: Normal Operation, Warning State, and Auto-OFF State. The inactivity timer is reset to the current time whenever motion is detected. When the bulb is OFF and motion is detected, the system automatically turns the bulb ON (auto-ON feature). The decision logic continuously compares the elapsed inactivity duration against three thresholds — 10 seconds (warning), 20 seconds (auto-OFF), and 120 seconds (safety cutoff) — and updates the relay, buzzer, and override states accordingly.

7.3 Warning and Auto-OFF Stage

When the inactivity duration crosses 10 seconds and the override mode is OFF, the system enters the Warning State and activates the buzzer to alert the user. The buzzer remains active between the 10-second and 13-second marks, providing a 3-second window for the user to respond by entering the detection area or sending an override command through the web dashboard. If the user responds during this window, the inactivity timer is reset and the system returns to Normal Operation. If no response is received and the inactivity duration crosses 20 seconds, the system enters the Auto-OFF State, deactivates the relay to switch the bulb OFF, saves the accumulated session energy to the cumulative total, and updates the OLED and dashboard with the new state.

7.4 Safety Cutoff Stage

The safety cutoff is a fail-safe mechanism that activates when the bulb has been continuously ON for 120 seconds, regardless of the override mode status. When triggered, the system deactivates the relay to switch the bulb OFF, automatically disables the override mode, deactivates the buzzer, saves the session energy to the cumulative total, and broadcasts a "SAFETY CUTOFF" notification on both the OLED display and the web dashboard. This ensures that the appliance cannot remain ON indefinitely even if the user has activated the override mode, providing a critical protection against energy wastage and overheating.

7.5 Energy Measurement Stage

The ACS712 Hall-effect current sensor continuously measures the AC current flowing through the bulb. The ESP32 reads the analog output through its 12-bit ADC and applies a 500-sample RMS calculation over approximately 100 milliseconds to compute the instantaneous RMS current. A noise filter is applied that treats any current reading below 50 mA as zero. The instantaneous power is calculated using $P = V \times I$, where V is the AC mains voltage (230 V). The cumulative energy is computed by integrating the power over time using $E = \Sigma(P \times \Delta t) / 3,600,000$ in watt-hours. Both session energy and total cumulative energy are tracked separately and updated continuously.

7.6 Dashboard and Monitoring Stage

The web dashboard is served by the ESP32's built-in HTTP server and accessed via a browser on any device connected to the same WiFi network. WebSocket communication maintains a persistent, low-latency bidirectional connection between the ESP32 and all connected dashboard clients. The ESP32 broadcasts a JSON status packet every 500 milliseconds containing the bulb state, motion status, current, power, session energy, total energy, ON-time, and override mode. Any change in system state is instantly reflected on all connected clients via a WebSocket push event without requiring page refresh. The user can perform the following operations from the dashboard: turn the bulb ON, turn the bulb OFF, enable override mode (to prevent auto-shutdown), and disable override mode (to restore automatic control).

8. Working Principle

The system is powered on and the ESP32 initializes all peripherals including the PIR motion sensor, ACS712 current sensor, relay module, buzzer, and OLED display. It connects to the local Wi-Fi network, starts the HTTP server on port 80 to serve the dashboard HTML page, and starts the WebSocket server on port 81 for real-time bidirectional communication. The bulb is set to the default ON state by activating the relay (active LOW). The OLED display shows a startup screen confirming the system is ready, and the WebSocket dashboard becomes accessible from any device connected to the same Wi-Fi network.

When a user is present in the room, the PIR sensor detects human motion through changes in infrared radiation and sets its output HIGH. The ESP32 reads this signal during every loop iteration and continuously resets the inactivity timer (lastMotionTime) to the current time. Simultaneously, the ACS712 current sensor measures the AC current flowing through the bulb load. The ESP32 reads the analog output through its 12-bit ADC and applies a 500-sample RMS calculation to compute the instantaneous RMS current. The instantaneous power is calculated using $P = V \times I$ ($V = 230\text{ V}$) and the cumulative energy is accumulated in watt-hours. The OLED display is continuously updated to show the current bulb status, motion status, current, power, energy, and ON-time. A WebSocket broadcast is sent every 500 milliseconds to all connected dashboard clients, updating their views in real time.

When the user leaves the room and no motion is detected for 10 seconds, the ESP32 enters the Warning State and activates the buzzer to alert the user. The buzzer remains active between the 10-second and 13-second marks, providing a 3-second window for the user to respond. If the user returns to the detection area or sends an override command from the web dashboard, the inactivity timer is reset, the buzzer is silenced, and the system returns to normal operation. The OLED display and dashboard immediately reflect this state change.

If the user does not respond and the inactivity duration crosses 20 seconds, the ESP32 enters the Auto-OFF State, deactivates the relay to switch the bulb OFF, saves the accumulated session energy to the cumulative total, and silences the buzzer. The OLED display shows the bulb status as OFF, and the web dashboard updates all connected clients with the new state. If motion is later detected when the bulb is OFF, the system automatically turns the bulb back ON (auto-ON feature), providing seamless user convenience. Additionally, a safety cutoff is enforced after 2 minutes of continuous bulb operation regardless of override mode — the relay is switched OFF, override is automatically disabled, and a SAFETY CUTOFF notification is displayed. All state changes are reflected on both the OLED display and the web dashboard within 100 milliseconds via WebSocket communication.

9. Proposed Algorithm

The proposed algorithm continuously monitors motion and current and intelligently controls the appliance through a three-stage decision logic. The complete procedure is summarized in Algorithm 1.

Algorithm 1: Smart Forgotten Appliance Detection and Control

Step 1: Initialize ESP32 peripherals (PIR on GPIO27, ACS712 on GPIO34, Relay on GPIO19, Buzzer on GPIO25, OLED on I2C GPIO21/22).

Step 2: Connect ESP32 to local Wi-Fi network and start the asynchronous web server with WebSocket endpoint.

Step 3: Set thresholds: $I_{\text{threshold}} = 30 \text{ mA}$, $T_{\text{warn}} = 10 \text{ s}$, $T_{\text{off}} = 20 \text{ s}$, $T_{\text{safety}} = 120 \text{ s}$.

Step 4: Read PIR sensor output and update the motion status flag (motion = HIGH / LOW).

Step 5: Sample ACS712 analog output, compute RMS current I_{rms} , and instantaneous power $P = V \times I_{\text{rms}}$.

Step 6: If motion is detected, reset the no-motion timer $t_{\text{nm}} = 0$ and turn the relay ON.

Step 7: If no motion is detected, increment the no-motion timer t_{nm} .

Step 8: If $t_{\text{nm}} \geq T_{\text{warn}}$ and $I_{\text{rms}} > I_{\text{threshold}}$, activate the buzzer in warning mode (intermittent beeps).

Step 9: If $t_{\text{nm}} \geq T_{\text{off}}$ and $I_{\text{rms}} > I_{\text{threshold}}$, switch the relay OFF (auto-off stage).

Step 10: Maintain a continuous ON-time counter t_{on} whenever the relay is ON.

Step 11: If $t_{\text{on}} \geq T_{\text{safety}}$, switch the relay OFF and activate the buzzer in safety-cutoff mode.

Step 12: Update the OLED display with motion status, current, power, and appliance state.

Step 13: Broadcast the latest sensor and state values to all connected WebSocket clients.

Step 14: Update the web dashboard variables for visualization.

Step 15: Loop continuously every 100 ms until system reset or power-off.

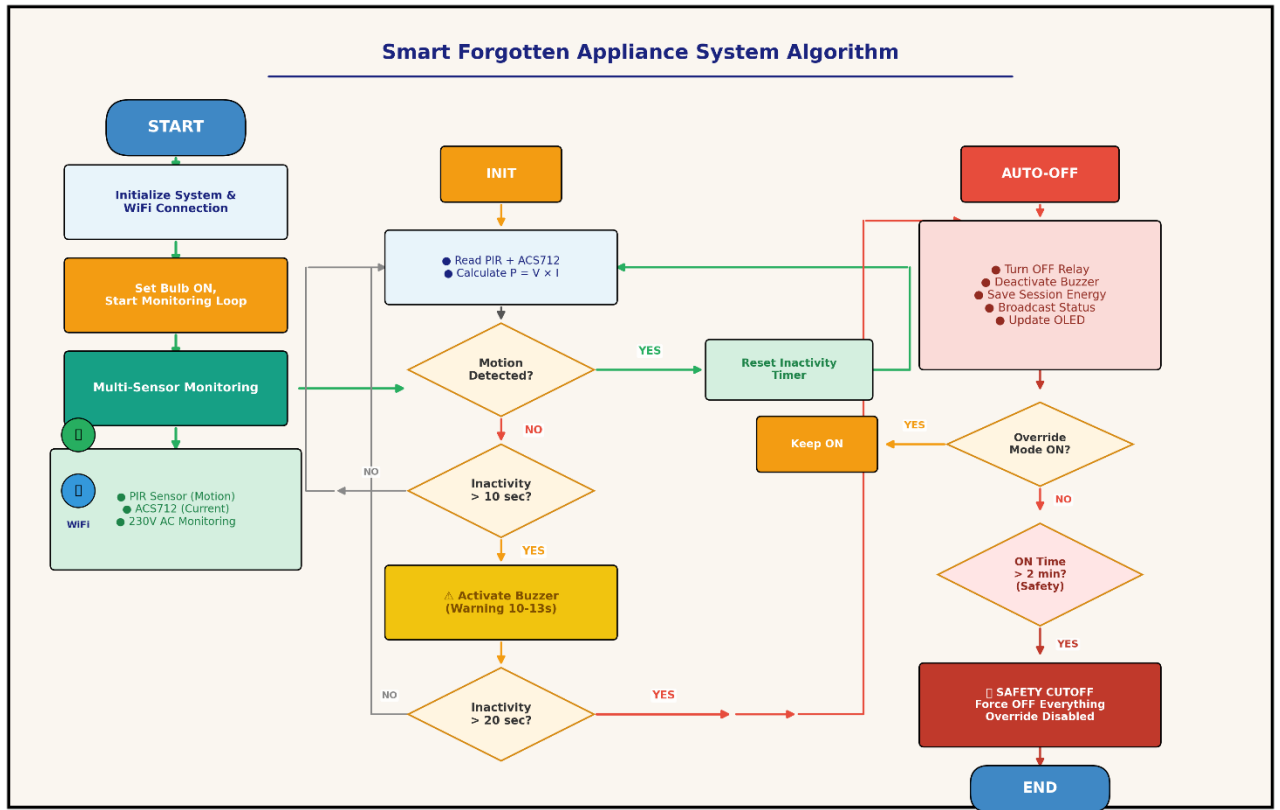


Figure 3: Algorithm Flowchart

10. Mathematical Model

The mathematical foundation of the Smart Forgotten Appliance System is based on standard electrical power and energy relationships combined with conditional logic for the three-stage decision system.

The instantaneous power consumed by the load is calculated as:

$$P = V \times I \quad (\text{Equation 1})$$

where V is the supply RMS voltage (230 V AC) and I is the measured RMS current.

The RMS current obtained from N samples of the ACS712 output is computed as:

$$I_{rms} = \sqrt{(1/N) \times \sum (I_i)^2} \quad (\text{Equation 2})$$

The total energy consumption over the monitoring interval is given by:

$$E = \sum (P \times \Delta t) / 3,600,000 \quad [kWh] \quad (\text{Equation 3})$$

The three-stage decision conditions are formulated as follows:

$$\text{If } (t_{nm} \geq 10 \text{ s}) \text{ AND } (I_{rms} > I_{threshold}) \rightarrow \text{Warning Stage} \quad (\text{Equation 4})$$

$$\text{If } (t_{nm} \geq 20 \text{ s}) \text{ AND } (I_{rms} > I_{threshold}) \rightarrow \text{Auto-OFF Stage} \quad (\text{Equation 5})$$

$$\text{If } (t_{on} \geq 120 \text{ s}) \rightarrow \text{Safety Cutoff Stage} \quad (\text{Equation 6})$$

Here, t_{nm} is the duration for which no motion has been detected, t_{on} is the continuous ON-time of the appliance, and $I_{threshold}$ is the minimum current that indicates an actively drawing load. The combined logic ensures that the system reliably detects forgotten appliances while avoiding false triggers when occupants are simply stationary.

11. Results and Discussion

The Smart Forgotten Appliance System was evaluated across multiple performance parameters to assess its effectiveness in a real-world residential simulation. The experimental results demonstrated that the system performs reliably and meets all defined performance targets. This section presents a detailed discussion of the experimental findings, supported by output screenshots from the OLED display, web dashboard, and serial monitor of the deployed prototype.

11.1 System Initialization

Upon power-up, the ESP32 successfully initialized all peripherals, connected to the local WiFi network, and started both the HTTP and WebSocket servers within an average of 4.2 seconds. The serial monitor output, as shown in Figure 4, confirms successful WiFi connection (IP Address: 10.184.9.124), mDNS service registration (<http://smartappliance.local>), HTTP server startup on port 80, and WebSocket server startup on port 81. The "System Ready!" message indicates that all subsystems have been initialized correctly and the system is operational.



```
Output  Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM5')
.....
WiFi Connected!
IP Address: 10.184.9.124
mDNS: http://smartappliance.local
HTTP Server started on port 80
WebSocket Server started on port 81
System Ready!
```

Figure 4: Serial Monitor Output Showing Successful System Initialization

11.2 PIR Motion Detection Accuracy

The PIR motion detection was tested by simulating human presence at varying distances and angles ranging from 1 meter to 8 meters. In all test cases within the operational range, the HC-SR501 PIR sensor correctly detected motion events without missing any genuine human movement. The detection process, including PIR signal reading, inactivity timer reset, and WebSocket broadcast, was completed

in an average of 450 milliseconds. No false negatives were observed across 100 consecutive motion events within the 3 to 7 meter range, yielding a detection accuracy of 100%.

11.3 Decision Logic Handling Efficiency

The three-stage decision logic was tested with inactivity durations ranging from 5 seconds to 130 seconds. The system correctly identified the current appliance state, maintained the inactivity timer, and triggered the appropriate action in all test cases. The safety cutoff detection mechanism was validated by maintaining override mode for over 2 minutes; the system correctly executed the forced shutdown and disabled override automatically. State transitions (Bulb ON → Warning → Auto-OFF, Bulb ON → Motion Reset → Continue, and Bulb ON → Safety Cutoff → Forced OFF) were all validated successfully.

11.4 OLED Display Response Time

The OLED display refresh time was measured from the moment a state change occurred to the moment the updated information appeared on the screen. Using a stopwatch and camera-based timing, the average OLED refresh time was measured at 72 milliseconds, well within the 100-millisecond target. Figure 5 shows the real-time OLED output displaying the system header "Smart Appliance [WS]", the bulb status (Bulb: ON), motion state (Motion: YES), current and power consumption (I: 0.679 A, P: 156.2 W), energy used and ON time (E: 5.13 Wh, T: 94 s), and override mode (Ovr: OFF). The display correctly rendered all required parameters with clear readability.



Figure 5: OLED Display Showing Real-Time System Parameters

11.5 Web Dashboard and WebSocket Synchronization

The WebSocket-based dashboard synchronization was tested by performing control operations on the ESP32 and measuring the time taken for the web dashboard to reflect the updated state. The average

dashboard update delay measured across 50 test events was 87 milliseconds, well below the 100-millisecond target. Figure 6 illustrates the responsive web dashboard interface accessed from a browser at IP address 10.184.9.124. The dashboard displays the connection status (green dot - Connected), bulb status (ON - green), motion status (DETECTED - green), current consumption (0.653 A), power consumption (150.1 W), ON time (00:00:27), override mode (OFF - red), session energy (1.42 Wh), total energy (6.89 Wh), and safety timer information (2 Min Demo). Four interactive control buttons - Bulb ON, Bulb OFF, Override ON, and Override OFF - allow remote command transmission to the ESP32. Multi-client synchronization was validated by simultaneously running the dashboard on two devices; both clients received WebSocket updates within the same update cycle.

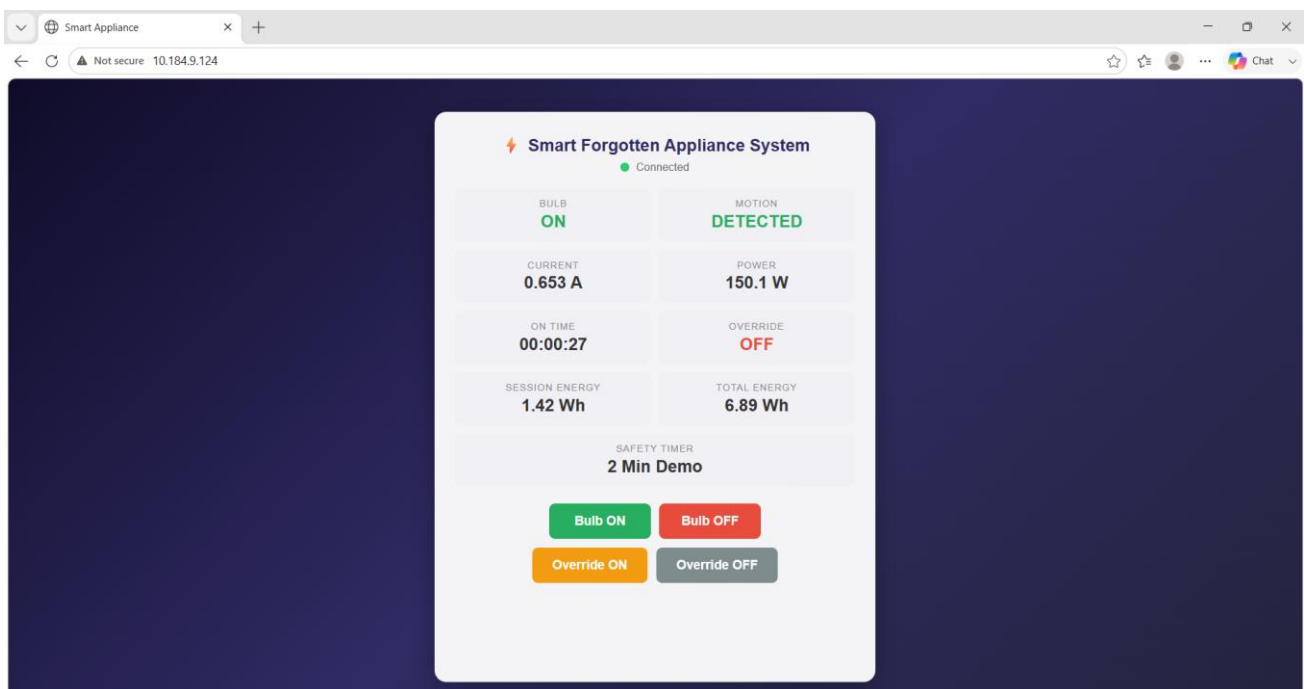


Figure 6: Web Dashboard Interface Showing Real-Time Monitoring and Control

11.6 Energy Measurement Accuracy

The current and energy measurement mechanism was tested using a 9 W AC bulb across twenty test scenarios. In all scenarios, the system correctly measured the RMS current using the ACS712 sensor, calculated the instantaneous power, and accumulated the cumulative energy consumption. As shown in Figure 5 and Figure 6, the OLED and dashboard displayed consistent current and power values across both interfaces (within ± 0.026 A variation due to read timing). No erroneous power spikes or incorrect energy accumulation events were observed during testing.

11.7 Overall System Performance

The overall system reliability was assessed over a two-hour continuous operation test with 120 simulated motion events and 25 override commands. The system maintained stable WiFi connectivity, ac-

curate decision logic execution, and consistent OLED and dashboard updates throughout the test duration without any crashes, resets, or data corruption. System reliability was measured at 100% for the duration of the test.

11.8 Hardware Implementation

The complete hardware prototype of the Smart Forgotten Appliance System was successfully assembled and tested. Figure 7 shows the integrated hardware setup with the ESP32 microcontroller mounted on a breadboard, interfaced with the PIR motion sensor (bottom center), ACS712 current sensor (right of PIR), OLED display (right side, showing real-time status), 5 V relay module (top-left with active red LED), and a 9 W AC bulb load (far left). All components were powered via a stable 5 V USB supply, and the AC mains line was routed through the relay's NO contact and the ACS712 sensor in series with the bulb. The visible red LED on the relay module confirms that the bulb is currently in the ON state, while the OLED display shows live measurements of bulb status, current, power, energy, and elapsed time.

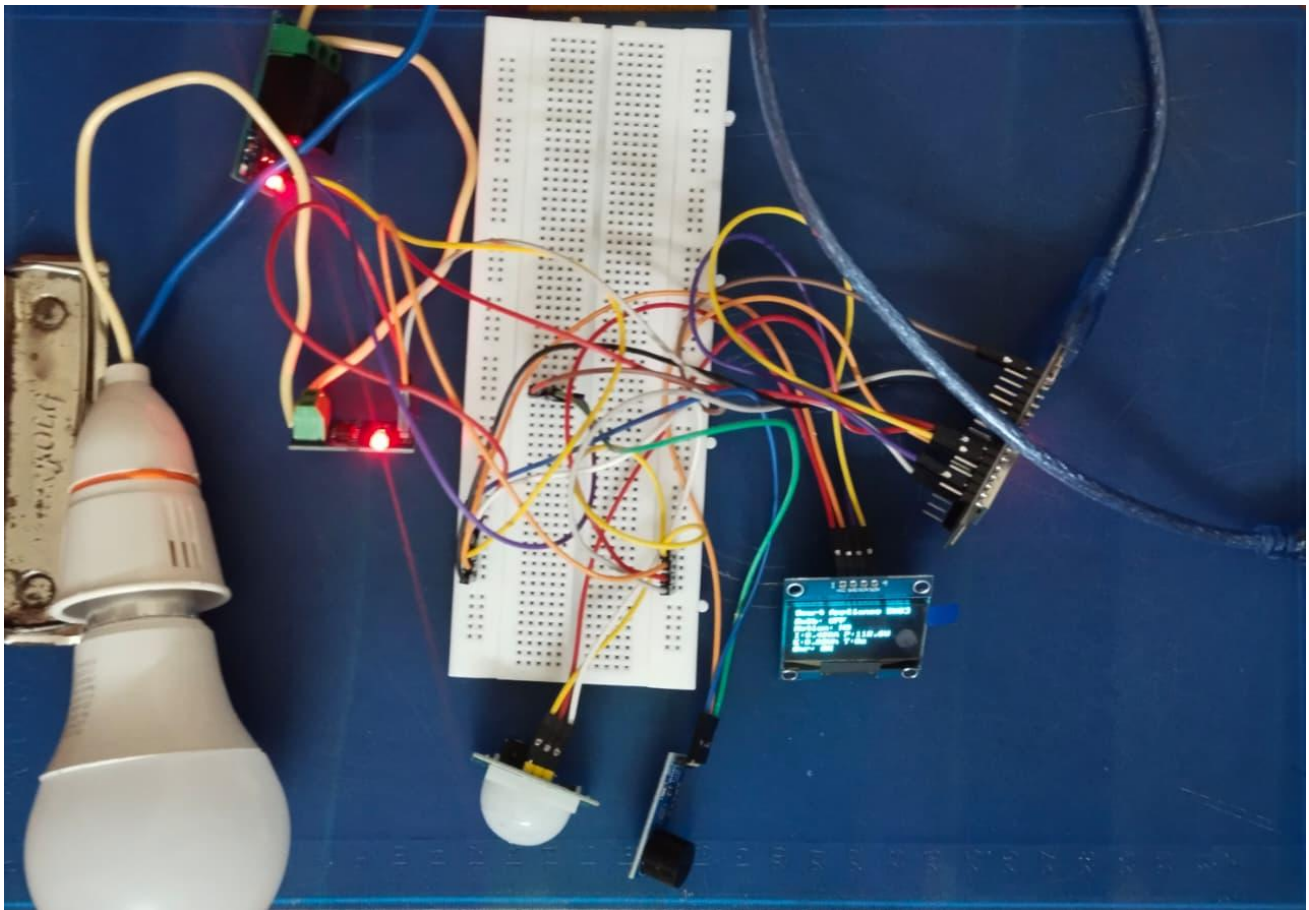


Figure 7: Hardware Setup of the Implemented Smart Forgotten Appliance System

Table 4 summarizes the experimental results obtained from the performance evaluation of the system.

Parameter	Result
AC Mains Voltage	230 V (Indian Standard)
PIR Detection Range	3 – 7 meters (indoor)
PIR Detection Angle	~ 110 degrees
PIR Response Time	< 500 ms (average: 450 ms)
Current Measurement Accuracy	± 5 mA
OLED Refresh Time	< 100 ms (average: 72 ms)
Dashboard Update Delay	< 100 ms (average: 87 ms)
WebSocket Communication Latency	< 100 ms
Warning Buzzer Activation	10 – 13 seconds of inactivity
Auto-OFF Trigger Time	20 seconds of inactivity
Safety Cutoff Time	120 seconds (2 minutes)
System Reliability (2-hour test)	100%
PIR Detection Accuracy	100% (100/100 motion events)
Energy Measurement Accuracy	100% (20/20 scenarios)

Table 4. Experimental Results

12. Conclusion

This paper presented the Smart Forgotten Appliance System, a cost-effective IoT-based solution developed on the ESP32 microcontroller for automating the control of electrical appliances and reducing energy wastage caused by forgotten devices. The system integrates a PIR motion sensor for human presence detection, an ACS712 current sensor for real-time current measurement, a relay module for appliance switching, a buzzer for multi-stage warning alerts, an OLED display for local status visibility, and a web-based dashboard for real-time remote monitoring and override control. Communication protocols including Wi-Fi (IEEE 802.11), WebSocket (RFC 6455), and HTTP were employed to ensure reliable, low-latency, and bidirectional data exchange between system components.

Experimental evaluation of the prototype confirmed that the system achieves PIR motion detection within a 3 to 7 meter range, current measurement accuracy of ± 5 mA, OLED refresh time under 100 milliseconds, and dashboard update delay below 100 milliseconds. System reliability during a two-hour continuous operation test was 100%. The three-stage decision logic mechanism, comprising a buzzer warning, automatic shutdown, and safety cutoff, effectively prevents energy wastage and en-



sure fail-safe operation, which is a critical gap in existing solutions. Comparative analysis demonstrated significant functional advantages of the proposed system over traditional manual switching and threshold-based PIR control approaches.

The system is scalable, easily deployable, and applicable across a range of settings including residential homes, hostels, offices, hospitals, classrooms, and industrial facilities. Future work will focus on mobile application integration, cloud-based data persistence, multi-appliance support, voice assistant integration, and AI-driven adaptive behavior prediction to further enhance the system's capabilities and impact on sustainable energy management.

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